

EDIM — Azure Development Deployment Plans

This document presents two low-cost Azure configurations for the development review phase of the EDIM energy-development modeling platform. The objective is to put the model runtime and frontend online for stakeholder review at the lowest reasonable cost, with the ability to configure and execute live model.

On Spot Virtual Machines: Azure Spot VMs offer unused datacenter capacity at discounts of 60–80% versus pay-as-you-go pricing. The trade-off is that Azure may evict the VM with 30 seconds' notice if capacity is needed elsewhere. For development workloads where occasional job loss is acceptable, Spot pricing provides the lowest possible compute cost. In this document, D4as v6 Spot is listed as an alternative for the model execution VM — at ~\$28/mo versus ~\$66/mo reserved. The **B4ps v2** option (1-year reserved) is recommended when predictable availability matters more than absolute minimum cost.

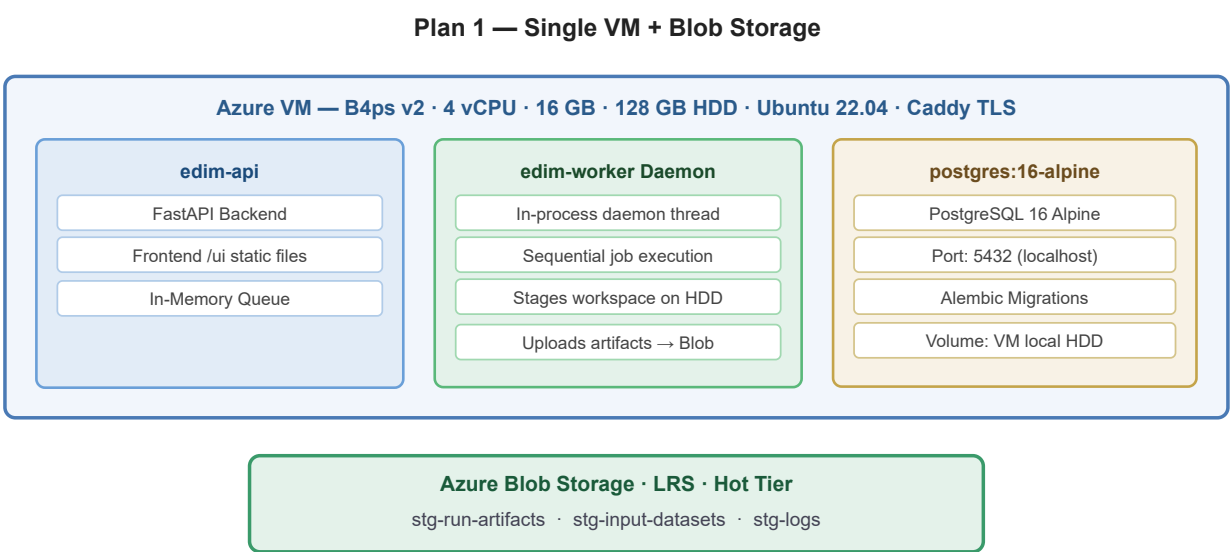
Pricing note: All prices in this document are rough estimates based on Azure retail pricing (West Europe, USD) as of June 2026. Actual costs may vary depending on region, currency exchange rates, reservation terms, and Microsoft pricing changes.

1. Plan 1: Single VM — Co-Located with Blob Storage External

The cheapest possible configuration. Ideal if the objective is to obtain stakeholder validation quickly and provide an environment for testing new implementations. Much faster to set up — a single VM, three Docker containers, no extra Azure services. No user authentication: the system uses the same test-user header shim as the offline proof of concept. Reviewers access the platform with pre-seeded identities.

1.1 Architecture

One VM runs all workloads. The API process hosts a background worker thread for sequential model execution. PostgreSQL runs alongside as a container. Only Blob Storage is external.



1.2 Resource Allocation

Component	VM	Resources
API + Frontend	Same VM	FastAPI + <code>/ui</code> static files + Caddy TLS
Worker Thread	Daemon in API process	Sequential subprocess execution. In-memory queue.
PostgreSQL	Container, same VM	<code>postgres:16-alpine</code> , port 5432, local HDD volume
Blob Storage	External Azure	Artifacts, datasets, logs via Managed Identity

1.3 Azure Resources

Resource	Dev SKU	Specs	Monthly
VM	<code>Standard_B4ps_v2</code>	4 vCPU, 16 GB	\$65.54 (1yr reserved)
— alternative	<code>Standard_D4as_v6</code> Spot	4 vCPU, 16 GB	\$27.85 (Spot; evictable)
Managed Disk	<code>S10</code> Standard HDD	128 GB	~\$6
Public IP	Standard static	1	\$4
Blob Storage	LRS Hot	~10 GB	\$2
Total			\$78 (B4ps v2) / \$40 (Spot)

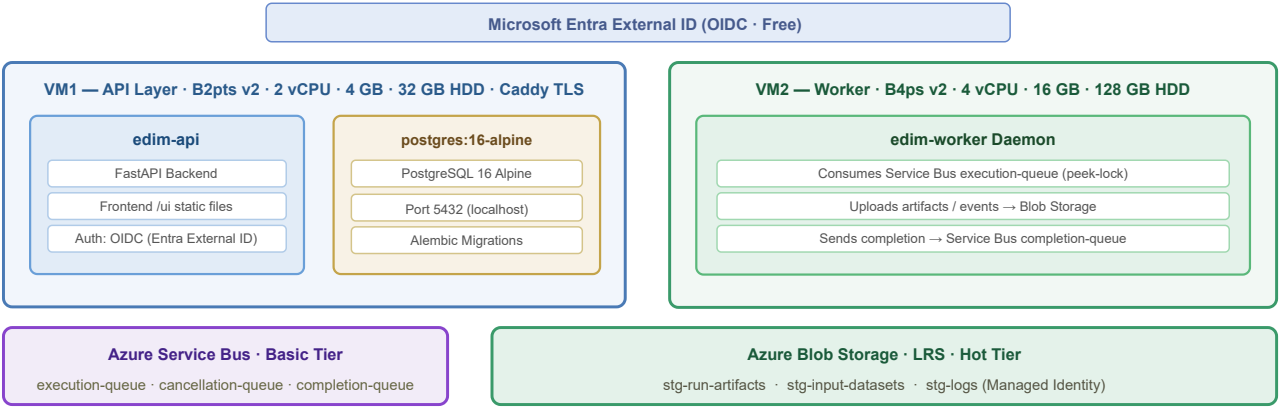
2. Plan 2: Two VMs — API+DB & Worker Separated, Service Bus Basic, Entra External ID

Aligned with the final production architecture. Worker isolation, durable messaging, and real user authentication via Microsoft Entra External ID. More complex to configure — two VMs, a Service Bus namespace, and an Entra tenant. The frontend needs basic login/logout process added. Infrastructure provisioning should begin as soon as possible to leave time for integration in order to comply with the deadline.

2.1 Architecture

VM1 (API Layer): API + frontend + PostgreSQL. **VM2 (Worker):** Model execution daemon only. Zero database access. **Service Bus Basic:** Three queues. First 13 million operations per month are free — effectively \$0 for dev. **Entra External ID:** OIDC auth, free for <50,000 MAU.

Plan 2 — Two VMs: API+DB & Worker · Service Bus · Entra External ID



2.2 Resource Allocation

Component	VM	Resources
API + Frontend	VM1	FastAPI + /ui + Caddy TLS
PostgreSQL	VM1 (container)	postgres:16-alpine, port 5432, local HDD
Worker Daemon	VM2	SB consumer → model CLI → Blob uploader
Service Bus	Azure Basic tier	execution-queue, cancellation-queue, completion-queue
Blob Storage	External Azure	Artifacts, datasets, logs
Entra External ID	Azure Free tier	OIDC token issuer, user sign-up/in

2.3 Azure Resources

Resource	Dev SKU	Specs	Monthly
VM1 (API+DB)	Standard_B2pts_v2	2 vCPU, 1 GB	\$4.11 (1yr reserved)
VM2 (Worker)	Standard_B4ps_v2	4 vCPU, 16 GB	\$65.54 (1yr reserved)
— alternative	Standard_D4as_v6 Spot	4 vCPU, 16 GB	\$27.85 (Spot)
Disk VM1	S4 Standard HDD	32 GB	~\$2
Disk VM2	S10 Standard HDD	128 GB	~\$6
Public IP (VM1 only)	Standard static	1	\$4
Service Bus	Basic tier	3 queues	\$0 (<13M ops free)
Blob Storage	LRS Hot	~10 GB	\$2
Entra External ID	Free tier	<50k MAU	\$0
Total			\$84 (B4ps v2) / \$46 (Spot)

3. Plan Comparison

	Plan 1	Plan 2
Monthly (B4ps v2)	\$78	\$84
Monthly (Spot)	\$40	\$46
API VM	Shared on worker VM	B2pts v2 (\$4.11)
Worker VM	B4ps v2 / D4as v6 Spot	B4ps v2 / D4as v6 Spot
Worker isolation	Daemon thread in API process	Separate VM, zero DB access
Queue	In-memory	Service Bus Basic (free)
Auth	X-EDIM-User-Id header	Entra External ID (OIDC)
Job durability	Lost on VM restart	SB retains messages; retries
Ops overhead	Low (1 VM)	Medium (2 VMs, SB, Entra)

4. VM SKU Selection Rationale

All prices sourced from [Azure Linux VM pricing](#) (West Europe, USD).

4.1 Development — Worker / Compute

SKU	vCPU	RAM	1yr Reserved	Spot	Why
B4ps v2	4	16 GB	\$65.54/mo	\$97.82/mo	Burstable B-series. CPU credits for bursty solver runs. Reliable.
D4as v6	4	16 GB	\$91.67/mo	\$27.85/mo	70% cheaper on Spot. Eviction risk.

4.2 Development — API Layer

SKU	vCPU	RAM	1yr Reserved	Why
B2pts v2	2	1 GB	\$4.11/mo	Cheapest 2 vCPU SKU. Sufficient for API + Caddy. PostgreSQL uses ~200 MB.

4.3 Storage — Standard HDD

SKU	Size	Monthly	Rationale
S4	32 GB	~\$2	API VM — OS + DB only
S10	128 GB	~\$6	Worker / single VM — Docker images, run workspaces

Sequential I/O. No premium storage needed. Run artifacts served from Blob, not local disk.

5. Production Target

The current baseline architecture (production-grade topology) costs approximately **\$163/mo**:

Layer	Azure Service	SKU	Monthly
Frontend	Static Web App	Free tier	\$0
API	App Service (Linux)	Basic B2 (2 vCPU, 3.5 GB)	\$25.55
Isolated Worker	VM	Standard_B4s_v2 (4 vCPU, 16 GB)	\$81.19 (1yr reserved)
Database	PostgreSQL Flexible Server	B1ms (1 vCPU, 2 GB)	\$12.41
Database storage	PostgreSQL Flexible Server	10 GB	\$1.15
Message Queue	Service Bus	Free Tier	\$0
Object Storage	Blob Storage	LRS	~\$3
Authentication	Entra External ID	Free (<50k MAU)	\$0
Total			~\$125/mo

All prices are billed upfront, year-based.

Production prices sourced from [App Service Linux](#), [PostgreSQL Flexible Server](#), [Static Web Apps](#), and [Linux VMs](#) (June 2026). Please note that the Worker VM could be insufficient for higher loads and users count, and more suitable (and costly) VMs could be necessary.